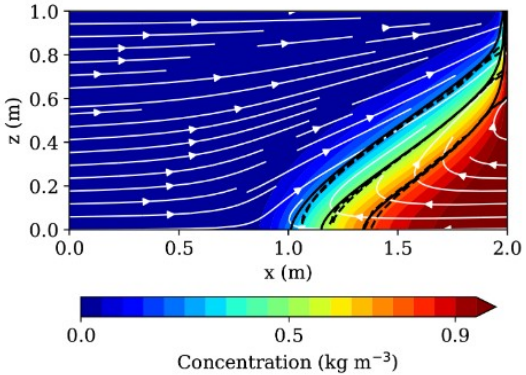
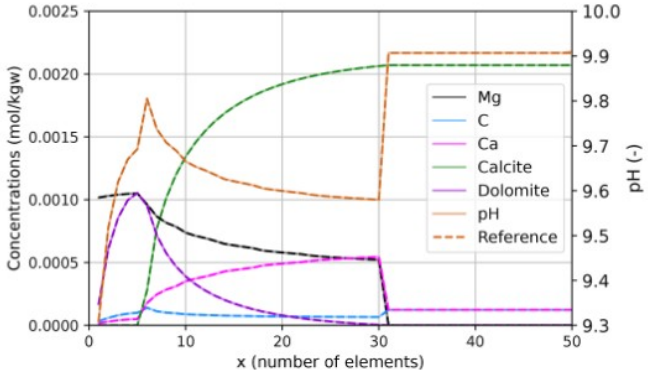
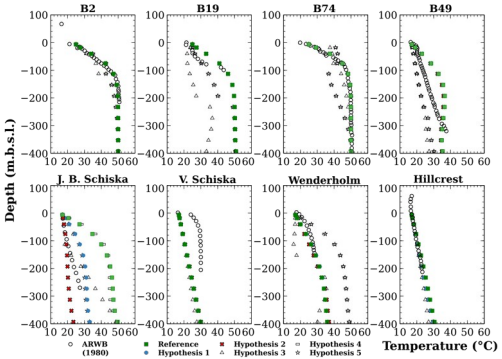
TRANSPORT Simulation Environment

Course simulations

Thomas Kempka
GFZ Helmholtz Centre for Geosciences / University of Potsdam
29/10/2026 - Potsdam







Course schedule focusses on simulations with increasing complexity and process integration

01 **Groundwater flow**

02 **Groundwater flow** with advective, conductive and dispersive transport of **heat**

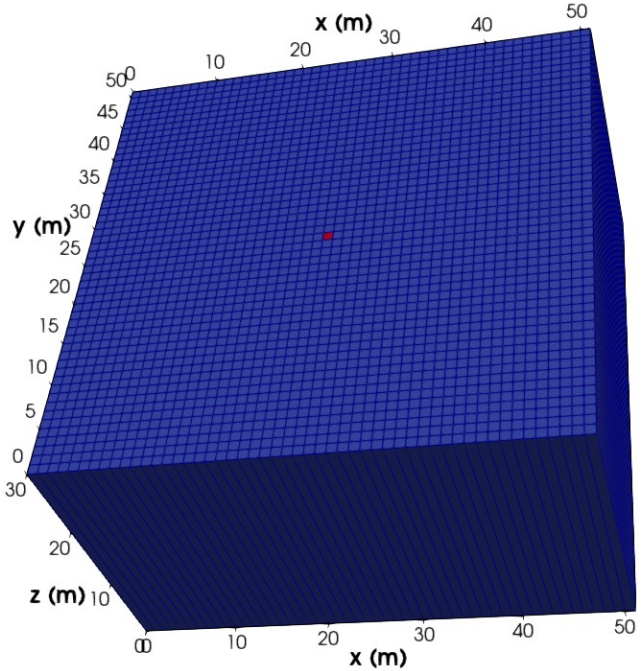
03 **Groundwater flow** with advective, conductive and dispersive transport of **heat** as well as advective, diffusive and dispersive transport of a **passive tracer**

04 **Density-driven groundwater flow** based on **chemical species** concentrations and transport of heat using **fluid equations of state**

05 **Groundwater flow** with transport of **heat** and **chemical species** considering **chemical reactions** by a coupling to a **surrogate model**

06 **Groundwater flow** with transport of **heat** and **chemical species** considering **chemical reactions** by a coupling to **PHREEQC**

2D geothermal reservoir model used as base case



- Model extent 51 m x 51 m x 30 m with 1 m x 1 m x 30 m grid discretisation
- Initial reservoir pressure is 15 MPa
- Injection well located at model centre with 50 $m^3\ h^{-1}$source term
- Lateral model boundaries are open (Dirichlet-type conditions)
- Top and bottom boundaries are closed (Neumann 'no-flow' type)
- Refer to specific config.py for parametrisation of each example
- Fluid EOS and chemical equlibria (later examples) derived from PHREEQC

Kühn, M. and Schneider, W. (2003) in Clauser, C. (ed.)
Numerical Simulation of Reactive Flow in Hot Aquifers. pp 253-262